

SHIMANOVSK, Russian Federation

WMO: 314420

Lat: 51.9868N

Lon: 127.6530E

Elev: 920

StdP: 14.21

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.3	-32.0	-41.1	0.5	-35.7	-36.9	0.7	-31.2	15.4	-4.9	13.8	-3.8	0.2	340	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	85.1	67.4	82.1	66.7	79.1	65.6	71.7	79.3	69.9	77.1	68.2	75.6	7.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	111.6	75.3	67.5	104.5	73.3	65.7	98.2	71.7	36.3	79.2	34.6	77.0	33.2	75.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.6	12.8	DB	-42.7	91.5	5.3	3.6	-46.5	94.1	-49.6	96.2	-52.6	98.2	-56.5	100.9	(4)
(5)			WB	-42.0	74.1	5.6	2.1	-46.0	75.6	-49.3	76.9	-52.4	78.1	-56.5	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	-10.6	-3.2	14.7	36.0	50.9	63.0	68.6	63.6	50.9	32.4	7.4	-9.5
	DBStd	29.70	8.97	9.10	11.11	8.35	7.46	6.27	4.91	5.68	8.08	9.54	11.99	9.26
	HDD50	8736	1880	1491	1094	427	78	2	0	1	87	551	1279	1846
	HDD65	12825	2345	1911	1559	869	443	108	20	93	425	1012	1729	2311
	CDD50	1621	0	0	0	8	105	391	577	423	113	4	0	0
	CDD65	235	0	0	0	0	5	47	132	50	1	0	0	0
	CDH74	2248	0	0	0	7	109	620	1058	421	32	1	0	0
	CDH80	539	0	0	0	1	23	175	270	69	1	0	0	0
Wind	WSAvg	5.1	3.8	4.3	5.7	6.9	6.5	5.0	4.5	4.6	5.4	5.8	4.7	4.2
Precipitation	PrecAvg	20.2	0.2	0.2	0.4	0.9	2.2	2.8	5.0	3.7	2.3	1.0	0.5	0.4
	PrecMax	33.1	0.8	0.8	2.1	3.5	4.7	5.4	11.5	8.9	4.4	3.6	1.5	0.7
	PrecMin	15.3	0.0	0.0	0.0	0.0	0.4	0.6	2.0	0.7	0.7	0.1	0.0	0.0
	PrecStd	3.9	0.2	0.2	0.4	0.8	1.1	1.4	2.6	1.6	1.1	0.7	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	26.6	47.8	72.8	83.3	90.6	91.2	85.1	77.5	64.9	40.1	22.2
		MCWB	13.7	21.3	37.3	51.6	60.1	68.1	69.4	69.2	61.8	51.0	33.0	19.1
	2%	DB	10.4	22.0	41.8	62.5	76.5	84.5	85.9	81.8	73.8	57.1	33.6	14.1
		MCWB	8.2	18.7	32.9	46.2	56.5	65.4	69.1	67.8	60.8	46.1	29.5	12.2
	5%	DB	6.8	16.9	37.0	56.4	71.5	80.7	82.7	78.6	69.3	51.6	29.6	8.4
		MCWB	5.1	13.9	29.9	43.3	53.9	63.8	68.4	66.2	58.1	42.8	26.3	6.7
	10%	DB	3.1	12.2	32.4	51.3	67.0	76.9	79.7	75.7	65.2	46.9	25.1	4.0
		MCWB	1.8	9.7	27.3	40.1	51.8	62.4	67.7	65.2	56.2	39.5	22.4	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	22.1	37.6	52.4	61.4	72.9	74.0	72.9	65.4	53.0	33.9	19.3
		MCDB	16.2	25.8	47.4	69.8	77.8	83.0	82.2	80.4	71.8	63.0	39.1	22.5
	2%	WB	8.6	18.7	33.3	47.4	58.5	68.1	72.5	70.5	62.7	47.4	30.1	12.1
		MCDB	10.5	22.3	39.9	60.8	72.4	79.0	80.2	77.0	70.2	54.8	33.6	13.7
	5%	WB	5.1	14.0	30.4	43.9	55.7	66.1	70.9	68.8	60.2	43.6	26.3	6.8
		MCDB	6.8	16.7	37.0	55.2	67.6	76.4	78.5	75.2	66.5	50.8	29.6	8.4
	10%	WB	1.8	9.8	27.2	40.7	53.3	64.2	69.3	66.9	57.9	40.1	22.4	2.8
		MCDB	3.1	12.1	32.3	50.4	64.1	73.8	76.6	73.0	64.0	46.2	25.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	21.2	25.0	23.0	21.1	22.7	22.1	18.9	19.8	21.7	19.6	19.6	19.7
		MCDBR	20.9	28.6	25.7	31.2	33.0	29.6	25.2	26.0	28.1	27.7	21.4	19.6
		MCWBR	19.0	25.0	18.2	18.2	16.0	12.8	10.7	12.6	14.7	17.4	16.8	18.1
	5% WB	MCDBR	20.9	28.0	23.9	28.8	27.0	24.0	19.1	19.4	22.2	24.7	20.8	19.4
		MCWBR	19.1	24.5	16.9	17.2	13.6	12.0	9.6	10.7	12.8	16.5	16.4	17.9
Clear-Sky Solar Irradiance	taub	0.259	0.293	0.370	0.551	0.621	0.502	0.474	0.417	0.389	0.380	0.308	0.264	
	taud	2.160	2.088	1.927	1.666	1.610	1.991	2.126	2.278	2.268	2.121	2.135	2.110	
	Ebn at Noon	234	256	252	216	209	242	247	256	249	218	206	205	
	Edh at Noon	22	32	47	70	78	54	46	37	34	32	23	20	
All-Sky Solar Radiation	RadAvg	463	840	1376	1630	1705	1917	1720	1508	1178	770	483	356	
	RadStd	16	23	63	169	130	206	154	102	92	49	23	18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page